User
:CinemasActivity
:LocationHelper
:SharedPrefManager
:MapView (OSMDroid)
:Polygon
:Marker
:RetrofitClient
:FoursquareApiService
:Call<FsqPlacesResponse>
domain: Cinema
:CinemaAdapter
:Intent

Scenario 1: Permissions, Map Setup & Foursquare Query

1 openNearbyCinemas()
2 checkPermissionsAndLoadCinemas()

alt Permission Missing
3 ActivityCompat.requestPermissions(ACCESS_FINE_LOCATION)
4 onRequestPermissionsResult(GRANTED)

5 getCinemaRadiusKm()
6 radiusKm (e.g., 15)
7 getCurrentLocation(context, callback)
8 onLocationResult(lat, lon)
9 centerMapAndDrawRadius(userPoint, radiusKm)
10 getController().setZoom(adaptiveZoom)
11 getController().setCenter(userPoint)
12 new Polygon(mapView)
13 setPoints(Polygon.pointsAsCircle(userPoint, radiusMeters))
14 setFillColor(0x262196F3) & setStrokeColor(0xFF2196F3)
15 getOverlays().add(circle)
16 new Marker(mapView)
17 setPosition(userPoint) & setIcon(ic_my_location)
18 getOverlays().add(userMarker)
19 getInstance().getFoursquareApiService()
20 foursquareApiService
21 searchNearbyCinemas(latLng, "cinema", "10024", radiusMeters, "DISTANCE", 20)
22 call
23 enqueue(Callback<FsqPlacesResponse>)
24 onResponse(call, response)
25 mapToDomainCinemas(response.body().getResults())

loop [each FsqPlaceDto]
26 new Cinema(fsqId, name, lat, lon)
27 setAddress(formattedAddress)
28 setDistanceMeters(distance)
29 cinema

30 plotCinemaMarkersOnMap(domainCinemas)

loop [each cinema]
31 new Marker(mapView)
32 setPosition(cinemaGeoPoint) & setIcon(ic_cinema_pin)
33 setTitle(name) & setSnippet(distance + address)
34 getOverlays().add(marker)

35 setCinemas(domainCinemas)
36 notifyDataSetChanged()
37 displaySplitView(interactiveMap + cinemaList)

Scenario 2: Synchronized List Selection & Navigation Intent

38 clickCinemaItem(cinema)
39 onCinemaSelectedFromList(cinema)
40 getController().animateTo(cinemaMarker.getPosition())
41 showInfoWindow()
42 displayPopupBubble(name, distance, address)
43 clickDirections(cinema)
44 openCinemaNavigation(cinema)
45 new Intent(ACTION_VIEW, "geo:lat,lon?q=CinemaName")
46 mapIntent
47 startActivity(mapIntent)
48 launchGoogleMapsNavigation(directionsRoute)

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